TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 24

5-MINUTE REVISION

Data Transformation

When ANOVA assumptions fail: log, square root, angular and Box-Cox transformations, and back-transformation.

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STAT 3202 · Chapter revision

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THE IDEA TO KEEP

Transform to address a model problem, not to force significance.

REMEMBER

Choose a transformation from the data scale and residual behaviour. Document it and interpret results on the appropriate scale.

APPLY IT

Log transformation turns multiplicative effects into additive ones, but log(0) is undefined. Any offset must be justified and reported.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 Why transform?

Used when ANOVA assumptions fail, typically **unequal error variances (heteroscedasticity)**, non-normal errors or non-additive (multiplicative) effects.

1. Transform the data
2. Run ANOVA and mean separation on the **transformed** scale
3. **Back-transform** means to the original units for reporting

Never pick a transformation just to get the conclusion you want.

2 Which transformation?

Data	Variance	Use
Counts (Poisson)	$\sigma^2 = \mu$ or $c^2\mu$	**Square root**
Binomial percentages	$\sigma^2 = p(1-p)/n$	**Angular** (arcsine)
SD ∝ mean; multiplicative effects	$\sigma^2 = c^2\mu^2$	**Logarithmic**

3 Logarithmic

Positive data

$$X = \log Y$$

With zeros

$$X = \log(Y+1)$$

Back-transform: $Y = \text{antilog}\, X$. Logs need positive inputs. Example in the book: weed counts.

4 Square root

Counts

$$X = \sqrt{Y}$$

Zeros present

$$X = \sqrt{Y+1}$$

Very small counts: $\sqrt{Y} + \sqrt{Y+1}$. Back-transform $\sqrt{Y+1}$ with $Y = X^2 - 1$. Insect, larval, defective and diseased-seed counts.

5 Angular (arcsine) transformation

Transform

$$X = \sin^{-1}\sqrt{p}, \quad p = \frac{Y}{100}$$

0% and 100%

$$p = \frac{1}{4n}, \quad p = 1 - \frac{1}{4n}$$

Back-transform

$$Y = 100\sin^2 X$$

For percentages from **binomial counts** (germination, mortality). The 30–70% guideline is approximate. Nutrient percentages (protein, carbohydrate) are not binomial and need not be transformed.

6 Box-Cox

A **family of power transformations** indexed by λ, for strictly positive data.

$\lambda = \frac{1}{2}$: square root · $\lambda = -1$: reciprocal · $\lambda \to 0$: log. The angular transformation is **not** in this family.

7 After transforming

- ✓ Check assumptions using the **residuals** of the fitted model.
- ✓ Use one transformation consistently for the whole dataset.
- ✓ CD on the transformed scale: $t_{\alpha/2,\,edf}\sqrt{2\,MSE/r}$ (RBD).
- ✓ Back-transformed means are **not** arithmetic means (e.g. squared root means are ≤ the original mean).

! Common mistakes

- ✗ Reporting transformed means as final results.
- ✗ Taking log of zero.
- ✗ Arcsine on non-binomial percentages.

? Fill in the blanks

1. Counts of insects per plant are usually analysed after a ____ transformation. **[square root]**
2. Percent germination out of 100 seeds calls for the ____ transformation. **[angular (arcsine)]**
3. When the SD rises in proportion to the mean, use the ____ transformation. **[logarithmic]**
4. A transformed mean √(Y + 1) = 3 back-transforms to ____. **[8]**
5. An angular value of 30° back-transforms to ____ %. **[25]**
6. Counts that include zeros are transformed as √(Y + ____). **[1]**